

On Predictive Diversity as a Function of Attentional Diversity

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Abstract

March and Simon positioned *attention* as a core concern of the Carnegie perspective because limited attention is one of the bounds on rationality. Drawing on the Carnegie-oriented “attention-based view” (ABV) of the firm and the wisdom of the crowd literature, this paper explores when an organization’s members’ diverse attentional foci facilitate successful organizational decision-making. The division of labor within organizations, and the cognitive drive to selectively apply attention according to one’s immediate tasks and goals, implies individuals have low-dimensional representations of their organization’s task domain, a form of bounded rationality. However, the wisdom of the crowd framework shows that decision makers with the right mixture of diverse representations can nonetheless collectively make accurate estimates. If attention within an organization can be structured to induce consistently diverse representations, organizations have the potential to “see the elephant” that its members cannot individually comprehend. To formalize this intuition, this paper presents a model of low-dimensional representations and the structure of attention within an organization to then analyze these factors as they relate to the overall potential to make correct decisions. The organization’s potential is always higher than any individual, but how much higher depends on the complexity of the decision-making environment, the number of decision makers, the size of individuals’ attention “budget”, and the particular distribution of attention among the decision makers. Although the model does not consider how decision-making processes are structured and the many other factors that could inhibit success, it clearly shows that limited individual attention, when channeled to diverse ends, creates a high potential for decision-making success.

Introduction

It is fitting that the scientific study of attention—the application of cognitive resources to particular mental processes or external stimuli—has had ups and downs in the amount of attention it receives.

Titchener called attention “the nerve of the whole psychological system” in 1908, but related research waned until the emergence of cognitive psychology in the 1960s. Among those interested in attention in this second era were Herbert Simon and James March, onetime colleagues at the Carnegie Institute of Technology, and founders of the now-eponymous Carnegie approach. Simon was broadly interested in decision-making and his notion of bounded rationality was rooted in the observation that individuals have limited cognitive resources and therefore can only pay attention to certain stimuli, issues, or decisions (Argote and Greve, 2007; Gavetti et al., 2007, 2012). This perspective firmly and explicitly placed attention as a core cognitive process involved in decision-making; in their introduction to the second edition of *Organizations*, March and Simon state that “although the central construct is decision-making, much of the theory developed in the book is less a theory of choice than a theory of attention” (March and Simon, 1993, p. 4). And yet, in spite of the general success of the Carnegie approach, attention fell off the research agenda until William Ocasio revived it around the turn of the century (Ocasio, 1997) through the attention-based view (ABV) of the firm.

The ABV is now a robust research program that applies the insights of research on attention to life within organizations. The primary claim around which much of this scholarship is built is that the attentional foci of decision makers will alter their strategic behavior (Ocasio, 1997). But, reflecting the capaciousness of attention as a set of cognitive processes¹, this literature still has “many unanswered questions on the relationship between organizational attention and strategic organization” (Ocasio et al., 2021). One theme missing from the ABV is the view of organizations as systems of distributed attention (*ibid*). This is surprising because the broader Carnegie approach sees organizations in part as distributed systems of information aggregation and processing (Simon, 1957; Stinchcombe, 1990; Puranam, 2018) and the distributed nature of attention within organizations should interact with those processes. To link these two perspectives, this paper explores the potential of organizations, as systems of distributed attention, to outperform any individual in decision-making tasks.

Moving in this direction requires a more positive interpretation of attention than is typical. Much of the literature in psychology has viewed limited attention as a bottleneck in information processing (Hommel et al., 2019), and the Carnegie perspective is no different (Porac and Tschang, 2013). Limited attention originates in our limited capacity for processing information (Broadbent, 1965) and what we do attend to often requires effort and directed control (Kahneman, 1973; Pashler et al., 2001). This means that stimuli are often excluded from further processing, and therefore limited attention is generally treated as a proximate cause of bounded rationality and misguided decision-making (Shiffrin and Schneider, 1977; Johnston and Dark, 1986). However, William James, as the first theorist of attention, saw the human attention system as often successfully allocating our limited processing capabilities toward important stimuli. He viewed attention as *preperception* that primes us to observe and process the most relevant stimuli effectively (James, 1890; Johnston and Dark, 1986).

¹So capacious that Hommel et al. (2019) titled their review piece “No one knows what attention is” as a rejection of William James’s claim that “everyone knows what attention is” .

This preperception side of attention has evolutionary roots (Hommel et al., 2019), but often originates in the socially derived constructs we use to interact with the world (Simon, 1957; Ocasio, 1997). From the welter of stimuli we could attend to, we have a strong tendency to attend to those that fulfill our specific tasks and goals, and match expectations and schemas².

This view suggests that when goals and tasks are circumscribed, selective attention can help us identify the most relevant stimuli and integrate the received information successfully. It does not, however, imply rationality in that circumscribed domain. Selective attention helps us avoid trying to process too much information, but the information we process might not be sufficient to resolve our uncertainty about a complex environment. This is where individuals embedded in organizations can do better. As a system of distributed attention, an organization can bring crucial structure to the parallel processing of information about the environment. By assigning individuals to specific tasks and domains—a division of labor—an organization channels, or selectively applies, individuals’ attention toward specific, locally relevant aspects of the broader environment. A marketing director should attend to consumer behavior and preferences to increase sales, while a head of engineering attends to things like design efficiency and cost control. The question becomes under what circumstances those individuals’ resulting representations of the broader environment can be reconciled for the purpose of successful organizational decision-making. If the relevant aspects of the environment are totally independent, recombination is straightforwardly plausible. But the environment is rarely so simple and often presents a system of interdependent choices (Siggelkow, 2011); a product’s design should depend on consumer behavior, and marketing choices should depend on the product’s design and cost. Thus, from an organization’s perspective, marketing and engineering goals and tasks should not be treated independently, even when the individuals in those roles selectively attend to their task domain and develop specialized representations of the organization’s environment.

Organizations get around the siloing of knowledge (and the shortcomings of individual decision-making) by using group processes to make decisions (Piezunka et al., 2022). Groups have an intelligence beyond that of the individuals in them, but the size of that factor *collective intelligence* depends on their particular collaborative processes (Riedl et al., 2021). Rather than addressing the wide range of possible decision-making processes, this paper uses a framework presaged in the early writings of the Carnegie tradition itself—the unstructured decision-making of a crowd. There is a long-standing interest in the phenomenon *wisdom of the crowd* because when the errors in judgment of individuals in the crowd are not correlated, the average of the individual estimates will have higher accuracy than most of those same individuals; even if individual errors are large, errors in different directions and magnitudes can cancel each other out and yield a highly accurate average estimate (Davis-Stober et al., 2014).

²As research on transformer neural networks has shown, learning to selectively attend to the most relevant stimuli is a crucial element of inference (Vaswani et al., 2017) and well-directed attention becomes the *solution* to the bottleneck of the mind’s (or neural network’s) ability to process information and discern meaningful regularities.

The power of the average of a group has long been used to argue for democracy and markets (Aristotle, 1967; Landemore, 2012; Hayek, 1945; Treynor, 1987; Page, 2007), but not all crowds are wise (i.e., highly accurate) because accuracy depends on the balance of errors within the group. The formal measure of the degree to which errors are balanced is known as *predictive diversity* (Hong and Page, 2004)—the mean squared error from the group average. Predictive diversity can be the result of noisy estimates, but it is also the consequence of *cognitive diversity* (Page, 2007; Hong and Page, 2009). In what follows, I develop a model to analyze when the distribution of attention within an organization leads to the predictive diversity that can result in accurate collective decisions.

Specifically, the model explores the frequency with which a group of decision makers within an organization can correctly choose which of two possible action alternatives has better performance. This particular outcome is more relevant than a measure of knowledge recombination due to the gap between judgments and decisions; within the judgment and decision-making literature, a decision is a choice based on the evaluation (judgment) of alternatives (Gilovich and Griffin, 2010). Because decisions can be correct without the judgments being accurate—this type of decision will be right as long as the rank ordering of judgment is right—assessing whether a binary choice decision is correct is as important as understanding the content and accuracy of individual judgments.

In what follows, I describe and extend the model of “low-dimensional representations” presented in Gavetti and Levinthal (2000). This model allows individuals to see only some of the dimensions of a high-dimensional problem space. Performance outcomes depend on the state of all dimensions, and so, through the lens of the low-dimensional representation, there is unexplained variation in performance. However, the authors show that these crude representations can provide actionable insight into the problem space. Although not discussed as a motivation for it, this is effectively a model of selective attention to the environment. I extend it by parameterizing an attention budget, or how many dimensions an individual can attend to before their understanding of the patterns in those dimensions starts to become more error-prone.

I also elaborate on the notion of an organization’s *attention matrix* as a distribution of attention that structures organizational members’ representation of the decision-making space³. On a high level, attention is directed toward the organization’s “principle premises” (Simon, 1957) and further guided by the economic and social structures in which it exists (Ocasio, 1997), but the division of labor also crucially channels attention by establishing circumscribed tasks and goals (Ocasio et al., 2021; Puranam, 2018). Within this framework, I analyze the consequences of these structures in terms of predictive diversity and the potential for an organization to make correct decisions. I focus on potential and not actual decisions because decision-making can be institutionalized in many different ways, and different processes vary in their ability to usefully extract latent potential. Considering a meaningful range of such processes is outside the scope of this paper.

³Cohen et al. (1976) introduces a similar concept as an extension the canonical “garbage can model” of organizational choice (Cohen et al., 1972).

The model yields several valuable insights. First, as we might expect, the probability of the organization making the correct decisions is always higher than the average probability that individual decision makers are correct. The gap narrows, however, as the decision-making environment becomes more complex. Second, there is often a non-linear relationship between the span of attention and accuracy when there is a cost associated with attention. And third, the precise distribution of attention can significantly alter the probability that the group is correct; attention matrices with more *attentional diversity* are better. Crucially, the attentional diversity creates more predictive diversity in the regions of the parameter space individuals are more likely to error in the first place.

This article is intended to be generative, not dispositive. In fact, to make progress relating an organization's allocation of attention to decision-making, it is necessary to set aside some the primary foci of the Carnegie tradition. The model is a behavioral theory of organizational decision-making at its core, but vital topics—search and learning, routines, emotions, ambiguity, luck, and appropriateness, just to name a few (Gavetti et al., 2007)—are not modeled. All of these considerations, in particular, the form of decision-making processes, will alter the final probability of the organization making correct decisions. However, this paper provides a baseline against which those results can be compared.

Modeling an organization's problem space

Organizational decision-making must be studied in relation to a particular problem space, and there are many possible spaces. All simple problems have the same general features, but complex problems have a yet-undetermined number of features that can affect how we make sense of them. To achieve generality in terms of the problem types, I follow the literature on performance landscapes and treat problems generically as a mapping from a set of N dimensions (about which decisions can be made) to an outcome space. If $|n_i|$ is the count of possible states of the i^{th} dimension, there are $|n_1| \times |n_2| \times \dots \times |n_N|$ different combinations, or overall states. For example, a sandwich shop might offer ten ingredients that can be included in a sandwich. We can model the ingredients as the dimensions and the possible states as being included or not. In this case, there are $2 \times 2 \times \dots \times 2 = 2^{10} = 1024$ combinations of ingredients, ranging from a simple cheese sandwich to a stacking cold cut combo. Because one can measure the distance between combinations based on how many ingredients or dimensions need to be changed to get from one to the other, combinations are often referred to as locations in a landscape of dimension N .

Choices around in which states the N different dimensions are becomes a problem in the Carnegie sense once the resulting combinations are associated with outcomes that can be compared or ranked in reference to a set of preferences or goals. One might prefer the cheese sandwich to the stacked cold cut combo and give it a higher outcome “score.” The characteristics of the problem space come from the nature of the mapping from combinations to outcomes. The mapping might originate in idiosyncratic tastes and budget constraints, just as it would in our hypothetical sandwich shop. But it may also be

dictated by fundamental forces, such as how the battery life of a laptop is dictated by the design of the components and their integration (Csaszar, 2018), or by market forces, in particular, the choices of consumers and competitors that establish the market value of a product.

From the perspective of a decision-maker, this problem space is latent. Organizations or individuals do not know the outcomes for all possible combinations. If they did, the rational thing to do would be simply select the highest scoring outcome; If someone tries all 1024 sandwiches, they can know which is best given their own taste profile. It is also latent in the sense that the space can expand overtime. The sandwich shop might someday add a vegan deli slice, which means there are now 2048 possible sandwich combinations. Meatless alternatives did not always exist, but the latent problem space can accommodate them in the sense that it is possible to conceive of the space as having always had 11 dimensions, but as a matter of history, the state of the 11th dimension was fixed as “not included.” Introducing vegan alternatives expands our notion of what is possible and, possibly, the outcome scores of combinations, but it does not fundamentally alter the representation of the problem.

This type of representation of an organization’s decision-making environment is useful for large strategic choices within organizations (Siggelkow, 2011); an organization is faced with the general problem of identifying high-scoring locations within the space of possible combinations and then “moving” there. An assumption of the search and learning literature within the Carnegie tradition, however, is that scores are revealed experientially. That is, the score of a location is only knowable once the actor has moved there and experienced the outcome. For this reason, a large body of research addresses the dynamics of effective experiential search (Baumann et al., 2019). In a departure from that literature, this paper does not address search. For reasons that will become clear below, I instead approach decision-making as the choice between two locations and then look at the probability of making the correct choice over *all* possible pairs of locations. Abstracting away from the dynamics of search facilitates a more precise analysis of the relationship between the distribution of attention and an organization’s ability to understand the problem space.

Given the flexibility of the mappings, there are infinite possible problem spaces. To maintain focus on the role of attention, I use a single, widely used class of problem spaces created by the *NK* fitness landscape model. This model for generating combinatorial landscapes has been used extensively for theorizing the processes of organizational search and learning in the Carnegie tradition because it facilitates variation in how challenging the problem is. The framework I present below can be applied to any class of problem space, but is an extension of a model that utilizes the *NK* landscape (Gavetti and Levinthal, 2000). Given this lineage and many readers familiarity with the *NK* model, I adopt it here. In the next section, I describe the *NK* landscape model before detailing the model of attention.

The NK model as a problem space

Due to its origins in evolutionary biology, the NK model is referred to as *fitness landscape*, the combinations as locations, and the outcomes as fitnesses. The N in the name NK is simply the number of attributes or dimensions of the problem space being modeled. The number of combinations grows exponentially in N , so small N landscapes (i.e., $N \leq 5$) represent problems for which all possible combinations can be easily enumerated. For example, when $N = 3$, the full set of binary combinations is $[0, 0, 0]$, $[1, 0, 0]$, $[0, 1, 0]$, $[0, 0, 1]$, $[1, 1, 0]$, $[1, 0, 1]$, $[0, 1, 1]$ and $[1, 1, 1]$. Increasing N leads to many more potential solutions to the problem. The number of dimensions is not the only source of problem complexity, however. The eponymous parameter K controls the degree to which the contributions to the outcome of a given dimension depend on the state of other dimensions. You might like a peanut butter and jelly sandwich just as much as a simple cheese sandwich, but a peanut butter, jelly, and cheese sandwich is likely to disappoint; there are obvious negative interdependencies of all three being included in the combination. The NK model creates interdependencies between attributes by defining the overall fitness of a location as the average fitness contributions of the N individual dimensions. When $K = 0$, the fitness contribution of each dimension depends only on the state of that dimension. There are two possible states for each dimension, and the fitness of each state is drawn from the uniform distribution $U([0, 1])$. This parameterization of the model is known as a “single-peaked” landscape because, for all locations save “the peak”, a dimension can change state to increase the overall fitness score. This is because one uniform draw is greater than the other, and switching to the associated state always increases the overall score. One can alter the state of each dimension one by one to “climb” to the highest overall fitness in less than $2N$ steps.

For $0 < K \leq N - 1$, the fitness contribution of the state of each dimension depends on the current state of K other dimensions. Rather than two draws—one for each state of the focal dimensions—there is a draw from $U([0, 1])$ for each of the 2^{K+1} combinations of states for the focal dimensions *and* those on which it depends. For example, when $N = 3$ and $K = 1$, the contribution of the first dimension could depend on its own state and that of the second dimension⁴. This means that there are four combinations to consider— $[0, 0]$, $[1, 0]$, $[0, 1]$, and $[1, 1]$ —each of which is associated with a distinct draw from $U([0, 1])$. If we change the state of the first dimension from 0 to 1 while the state of the second remains 0, the fitness sub-score of the first dimension is the draw associated with the combination $[1, 0]$. This calculation is repeated for all dimensions and the overall fitness score of the location is the average of these sub-scores.

Such interdependencies require that the sub-score an attribute needs to be recalculated whenever there is a change of state in the dimensions on which it depends. This makes it more likely that there are multiple combinations with higher scores than all their immediate neighbors, locations known as local optima or peaks. This quality means that increasing K also leads to more challenging search and

⁴The dependencies between dimensions can be hardcoded in an interdependency matrix and the precise pattern of interdependencies can alter the characteristics of the fitness landscape (Rivkin and Siggelkow, 2007).

learning problems, since a boundedly rational actor will be reluctant to move off of a local optimum, a scenario known as a competency trap (Levitt and March, 1988).

For defining representations below, it is useful to represent the latent problem space T as a $2^N \times (N + 1)$ matrix. The first N columns are each associated with one of the dimensions of the problem and the rows are associated with the 2^N possible combinations⁵. The last column is the outcome or fitness value for the combination identified by the rest of the row values. The matrix T below is an example of a single-peaked landscape for $N = 3$.

$$T_{8 \times 4} = \begin{bmatrix} 0 & 0 & 0 & .35 \\ 1 & 0 & 0 & .50 \\ 0 & 1 & 0 & .25 \\ 0 & 0 & 1 & .55 \\ 1 & 1 & 0 & .40 \\ 0 & 1 & 1 & .45 \\ 1 & 0 & 1 & .70 \\ 1 & 1 & 1 & .60 \end{bmatrix}$$

A model of attention and problem representation

Given T , the problem space in which decision-making occurs, we can now consider how the attention of decision makers can be allocated over this space. To do this, I extend the model presented in Gavetti and Levinthal (2000) (henceforth G&L), which gives actors *low dimensional* representations of the problem space. These representations are low-dimensional in the sense that they ignore some dimensions of the problem and are therefore simplified representations of reality. “As boundedly rational actors, we cannot envision the full set of alternatives available to us, nor can we completely specify the causal linkages between possible alternative actions and possible outcomes.” (p.177, G&L), but the need for forward-looking cognition is strong and, instead of blindly searching, we create and share simplified representations. G&L demonstrates that these low-dimensional representations, while far from complete, rational maps of the space, enable better outcomes than trail and error learning (i.e., experiential search).

G&L do not present selective attention to particular dimensions as the source of low-dimensional representations, but functionally the agents in the model are selectively attending to subsets of dimensions. The count of dimensions is parameter NI . If, for example, they attend to three attributes of a

⁵This combination count assumes that there are two possible states for each attribute. The same framework can be extended to any countable number of states or by discretizing values in a continuous space. Cognitively, the collapsing of infinite variety into a countable number of categories is common (Hannan et al., 2019).

problem with ten dimensions (i.e., $N=10$, $N_1=3$), they “see” only $2^3 = 8$ combinations of the possible 1024. All 1024 combinations remain distinct locations in the problem space and have distinct fitnesses associated with them, but the agent’s representation can only distinguish between combinations that differ in dimensions to which it attends.

Consider the problem space T from above and an agent who attends only to the first and second dimensions. The agent cannot distinguish between locations $[0, 0, 0]$ and $[0, 0, 1]$ in this case. Instead, the representation observes variation in the outcomes associated with the single perceived location $[0, 0, -]$. Because there are multiple fitness outcomes associated with the single perceived location, the low dimensional representation encodes the outcome at this location as the average of all the fitnesses associated *with the locations the representation cannot distinguish*. In our example, the representation’s fitness for $[0, 0, -]$ would be .45, the average fitness for $[0, 0, 0]$ and $[0, 0, 1]$, .35 and .55, respectively. In our 10 ingredient deli, a customer might not attend to whether the bread is toasted and some employees might not toast it. The customer’s evaluation of the stacked cold-cut sandwich would therefore be the average “score” of both toasted and untoasted versions of the sandwich.

This way of assigning fitnesses in a low-dimensional model merits two points of discussion. First, the use of the arithmetic mean can often be a very crude summary of the distribution of fitnesses within an NK landscape. Even when there are dozens of indistinguishable combinations for a focal perceived location, the mean could fail to capture the central tendency because the distribution need not be even unimodal, let alone normally distributed. The statistics of the marginal fitness distributions of the NK landscapes are quite complex (Weinberger, 1991), and cannot be easily embedded in a lower-dimensional space. For this reason, it is not obvious that G&L would find that such a crude representation as a mean can nonetheless improve search outcomes.

The second point is around the epistemic burdens of this model. The implementation of low-dimensional representations requires that the representation-generator function has access to the fitnesses of each of the indistinguishable points to calculate the mean value. However, access to such extensive knowledge seems to run counter to the present themes of selective attention and bounded rationality. G&L justifies this type of low-dimensional representation—which has some correspondence to the underlying truth of the problem—citing the work of Johnson-Laird (1995) and Weick (1990) on low dimensional representations. However, the Carnegie work on vicarious learning also supports it. Vicarious learning, or learning from the observation of others (Levitt and March, 1988; Miner and Mezias, 1996) is an important means through which individuals and organizations understand their environment and make choices. Vicarious learning is often cast in terms of imitation or diffusion, but at the lowest level, it is about knowing that others have adopted a practice or technology and observing the performance of the whole organization. Full-on imitation of complex strategies is often unsuccessful because the observer does not see the complete set of practices and technologies that gave rise to the performance (Rivkin, 2000), but the observation of others can still inform a low-dimensional representation.

Thus, the head of an IT department might observe other organizations adopting a new communication technology and determine that, on average, those organizations are doing better than their own and other non-adopting organizations. They are unable to distinguish between the organizations they are observing in the many ways they actually differ. Instead, they observe groups of organizations that do or do not have the focal dimension in the relevant state (i.e., is adopted) and then the range of outcomes associated with both groups. Whether or not the mean captures something about the contribution of the focal dimension, organizational decision-makers are implicitly, and often explicitly, tasked with this type of learning related to their attentional foci. Denrell (2003) rightly notes that there is likely to be selection bias in the observation set, but relevant information can also come through formal education, the business press, informal networks, and cultural learning.

So, while the model creates a crude approximation of the landscape by querying fitnesses at all indistinguishable points, my argument is not that selective attention provides us with similar information. Rather, when we selectively attend to some features of the problem, vicarious learning can get us to representations that have some rough correspondence with the true underlying landscape. The mean can have a rough correspondence and is convenient and straightforward. Yet, whether the mean is useful depends on the features of the landscape and the number of attributes the agent attends to. All things equal, the more dimensions attended to, the more accurate the low-dimensional representation. At the limit of attending to all dimensions, the representation is fully accurate. This is incongruent with the notion of attention, which is limited and selective due to our information processing constraints (Broadbent, 1965; Kahneman, 1973).

For this reason, I extend the G&L model to include an attention budget parameter, α . A budget captures the fact that, as a decision-maker attends to more features of the problem, they become less capable of meaningfully learning about the relationship between those features and performance outcomes. α is simply the number of dimensions that a decision maker can attend to before their representations become even cruder. In the analyses below, α is explored as an independent parameter because there is not yet a meaningful way to calibrate it; While much is known at this point about attention processes in the moment (Van Ede and Nobre, 2023), it is not clear how that relates to learning over an extended period of time.

Once an agent allocates attention to more dimensions than α , the fitnesses in their representation are drawn from a normal distribution with the mean of all fitnesses of the indistinguishable locations (as in the original G&L model) and a standard deviation $\sigma = \max\left(\frac{N1-\alpha}{N}, 0\right)$, where $N1$ is the count of dimensions attended to by the agent. Any values outside of the range of outcomes are rounded to the nearest endpoint of the range. The normal distribution is undefined for $\sigma = 0$ so the mean itself is used in the cases in which the decision maker attends to fewer dimensions than the budget allows.

For example, when $\alpha = 1$ and $N = 10$, a decision maker attending to two attributes has $\sigma = 1/10$. With a comparatively small standard deviation like this, an agent's fitness value for a location in their representation is likely to error slightly from the true mean of the indistinguishable locations. An

agent attending to six dimensions has fitnesses drawn from a distributions with $\sigma = 1/2$ and is wildly inaccurate in their assessment of the fitness of the location. With such large standard deviations, it is likely that the sampled value will be outside the range of possible fitness values, $[0,1]$. Rounding up or down to the extrema of the range therefore enforces a quasi-binary notion of fitness—good or bad—, which is the crudest possible low-dimensional representation. But even such crude representations can be informative because the mean fitness for indistinguishable points can still meaningfully influence whether the representation encodes the location as good or bad.

Attention as a driver of decision-making performance

G&L shows that this type of low-dimensional representation benefits search on NK landscapes. Because the representation is fully cognitive, it is searchable “offline”—without experiential exploration of the space. An agent identifies the highest value location within its representation and moves directly there. Doing so has an initial and long-running boost to the agent’s fitness score. Given sufficient time, an agent relying only on experiential search will mostly catch up, but the cumulative difference in performance is substantial.

The analyses that follow put aside the search problem to focus on decision-making across the entire problem space. Search processes are highly path-dependent and never exhaustive of the space. They therefore fail to capture the full extent the collective intelligence available to an organization with representationally diverse decision-makers. As with decision-making processes, the dynamics of search will be partially determinative of organization performance in practice, but this paper is first addressing the underlying potential. To capture the full extent of this potential, in the first analysis, I look at the frequency with which groups of decision-makers make the objectively correct choice for *all* possible decisions in a given problem space, conditional on their span of attention ($N1$), their attention budget (α), the size of group (M), and the complexity of the landscape (K). The second analysis below then introduces structure to the allocation of attention across the organization.

The focus of the first analysis is the relationship between the breadth of attention among individuals and decision-making success for individuals and groups. At one extreme, an agent who attends to all features of a problem and lacks any constraint on attention (i.e., an attention budget) can be perfectly rational and will make each decision correctly. But as attention becomes more costly, two questions arise. First, what is the impact on the percentage of decisions an individual makes correctly? It is clear that the percentage will decline, but the extent of decline is worth examining because of the second question; to what extent does diverse attention among the agents facilitate crowd wisdom and a high percentage of correct decisions? This is the core question of this paper.

The answer ultimately depends on the number of decision-makers and how their attention is structured at the level of the organization. The analysis of the role of attentional structure is in the next section, but throughout the analyses, I vary the number of decision-makers within the implied

organization via parameter M . The number of decision-makers available is a crucial determinant of the accuracy of the crowd because predictive diversity becomes more likely with more people. The maximum number of attentionally unique decision makers is determined by the number of dimensions agents attend to. When $N1=5$, there are 256 distinct possible allocations of attention, representing substantial representational diversity. When $N1=1$ or $N1=9$, there are only ten possible allocations, yet each set of ten representations is quite different and can have divergent effects on accuracy.

For both sets of analyses, $N = 10$. While real organizations often face higher dimensional problem spaces, many analyses of the NK landscape within the organizational learning literature focus on $N=10$. This problem space size also strikes a balance between possibly having numerous local peaks (for $K=9$, 70-80 peaks is typical), and yet have the total number of decisions be computationally tractable; with $N=10$, there are 1,048,576 decision pairs per landscape.

— INSERT FIGURE 1 ABOUT HERE —

Figure 1 reports the average performance results for a range of values of $N1$ and K —the extent of interdependencies between features. Each point is the average for the same simulation across 50 distinct landscapes because the stochastic process of constructing an NK landscape can result in each having distinct features that make it more or less challenging for agents to comprehend. For a given value of K , the 50 landscapes remain the same as the other parameters are varied⁶. When the number of decision makers is less than all possible (i.e., when $M=\text{“all”}$), the group versus individual comparison is for the same set of decision-makers randomly sampled from the set of all possible.

The percentage correct for individuals is simply the number of decisions for which the agent’s representation chooses the correct location when prompted to choose which of two locations had the higher fitness. Each agent considered all 1,048,576 pairs, including each location with itself, which are always coded as the agent making the correct choice. There are many instances in which the agent is unable to perceive locations as distinct when, in fact, they are. In those cases, the decision is effectively coded as correct 50% of the time—as if they flipped a fair coin—by contributing .5 to the count of correct answers.

The same basic logic applies for groups, but the first judgments of individuals are aggregated into a group decision. Each individual perceives a signed difference in fitness between the two locations, which is then averaged across all decision-makers in the group. This average is the group’s perceived difference, which then determines which location has the higher perceived fitness. Note that this outcome can be different from a majority vote because a large difference in one direction can effectively cancel out multiple small differences in the other direction.

This distinction is immediately relevant because it explains why the groups are highly accurate when $K=0$, as shown in the upper left facet of the figure. When the number of decision-makers is

⁶To maximize comparability, for given a single landscape, different sets of decision-makers are created for each value of $N1$. Thus, they are representing the same landscape, but in different ways as $N1$ increases.

greater than the number of dimensions in the problem space, the group is 100% accurate for $N1=1$. This fact provides insight into the mechanics of when crowds have the potential to perform well.

When there are no interdependencies between the dimensions for $K=0$, the crowd can correctly distinguish which of any two locations in the problem space has a higher fitness. Each decision maker is highly likely to correctly identify which state of the dimension to which they attend has the higher fitness, and, collectively, the decision makers have the knowledge of which location has better fitness. Each representation perceives distinctions between only two locations, but in reality there is a group of 512 different locations consistent with each location. Crucially, the difference in the average fitness between these groups is the same as the difference between the fitness contributions of the two states for the dimension to which the decision maker attends. To see this, assume that when the dimension attended to by a given decision maker is in state [1], the fitness contribution made by that dimension to the location consistent with that state is A . When the dimension is in state [0], the contribution is B . These contributions remain the same across all 512 locations within the respective groups the decision-maker can distinguish. Thus, the difference in *average* fitness between the group associated with state [1] and the group associated with [0] is $A - B$. Accordingly, when the dimension being in state [1] contributes more than being in state [0] (i.e., $A > B$), the averages fitness of the representation for the [1] group is $\frac{A-B}{N}$ greater than the [0] group, and the perceived difference always corresponds to the true difference for that dimension.

Other decision makers cannot perceive these differences on that dimension, so if the comparison is between two locations that differ on only one dimension, only the decision-maker who attends to that dimension knows the sign of the difference between the points. When that difference is averaged with the perceived differences of the others (i.e., 0), the average difference has the correct sign, and the group makes the correct choice every time. This logic can be extended to comparisons that involve changes in multiple dimensions. Those attending to the relevant dimensions know the contribution of each and the average of their differences is in fact the true difference, by the construction of the NK model. Even when averaged with the contributions of those decision-makers who don't make a distinction, the sign remains correct and the group will make the correct decision for any possible location comparison when there are as many unique decision-makers as there are dimensions to the problem. (Note that when $M=5$ and $N1=1$, there are five dimensions that are unattended to and accuracy drops to 72%.)

This is not true for individual decision makers. For the vast majority of comparisons, individuals do not perceive any difference and choose randomly, creating a baseline of 50% accuracy. When the two comparison locations differ on only the one dimension the decision-maker attends to, they are always correct. However, this is only a small percentage of all comparisons. For the rest, it is possible that the decision maker has the correct sign when the true sum of the differences on the other changed dimensions is less than the perceived difference. For example, consider a comparison that involves a change in three dimensions. If a decision-maker perceives a difference of +.12 on the dimension

they attend to, but the two other dimensions have differences of $-.14$ and $+.04$, the true difference is $+.02$, a positive difference. Our focal decision-maker also has a positive difference and will make the correct decision. The frequency with which this type of correct choice occurs depends on the K value and stochastic initialization of each NK landscape. Across 50 different landscapes and all decision makers, the frequency is roughly 7%, which is why the average individual accuracy for any number of decision makers turns out to be 57% for $N=0$ and $N1=1$.

As individuals attend to more dimensions, the general trend is for individual and group accuracy to increase. The exception is for groups of 20 and 10 decision-makers. When $N1=2$, there are 45 distinction allocations of two units of attention and the absence of 25 or 35 of them from these groups undercuts their accuracy relative to when $N1=1$, when there were only 10 possible allocations and those groups had complete coverage of the space. The dropoff caused by this change is not especially drastic. For $M=10$, the percentage of correct group decisions is approximately 86% despite the fact that the group has only 10 of the 45 possible representations. This demonstrates that there is a significant amount of predictive diversity in an arbitrarily chosen set of representations; individual errors in judgment offset each other in opposite directions so that the central tendency becomes accurate. Individual and group accuracy continues to improve as more units of attention are used.

All of the above, however, is conditional on the landscape lacking interdependencies (i.e, $K = 0$). As K increases and changes in one dimension depend on the states of other dimensions, the problem becomes harder and individual and group accuracy suffer. This is clearly visible across the remaining three panels of Figure 1. The greater the number of interdependencies, the lower the accuracy. Increasing the number of units of available attention still increases accuracy.

For the groups, having more decision makers provides surprisingly small marginal improvements. For example, for $K=3$, the increase in accuracy provided by including all 252 decision makers for $N1 = 5$ yields an accuracy improvement of approximately 5.5% relative to a group with a random sample of 5 decision makers (81% vs 75. 5%). For these same parameters, the difference between the average of the individuals and the group of 5 is 9.5%. This nonlinear relationship between the number of distinct decision makers and group accuracy indicates that adding predictive diversity becomes harder when decision makers are added at random. This follows from the simple fact that the probability of including a new representation that is relatively uncorrelated with the others becomes harder the more representations already included. However, it raises the question of whether providing more structure to the allocation of attention is a more effective means of adding predictive diversity, the topic of the next section.

The patterns in Figure 1 are predicated on there being no constraints on how many units of attention (i.e, $N1$) a decision maker can use before attention is spread too thin and their representations become noisier. This is because $\alpha = 9$, but the whole notion of attention is that we have a limited capacity to process information and we must be selective about the information we attend to. In Figure 2, α is varied to document the effect of an attention budget on the frequency of being correct. When $N1 \leq \alpha$,

the patterns are the same as in Figure 1. When $N1 > \alpha$, there is generally an immediate falloff in the percentage of correct decisions made by individuals. For $K=0$ and $K=3$, the dropoff continues as decision makers attend to more dimensions, but, interestingly, the accuracy percentage starts to increase again for $K=6$ and $K=9$. The percentage correct remains higher for lower K , but the return of the trend of the percent increasing as a function of $N1$ highlights a trade-off between depth and breadth of attention.

— INSERT FIGURE 2 ABOUT HERE —

This trade-off is more noticeable within the groups. Except for the parameters for which the complete group starts from a baseline of group rationality (i.e., $K=0$ and $\alpha=6$ and $\alpha=3$ and accuracy is at 100%), it is collectively better for the individuals within the complete group to attend to more dimensions than α . They are becoming less accurate on each individual dimension on average, but the information provided attending to one or more additional dimensions more than makes up for it. They are becoming less accurate individually, but more accurate collectively when all possible representations are included (i.e., $M = \text{“all”}$). When M is 5, 10, or 20, there is a substantial dropoff at $N1 = \alpha + 1$ and then, in some instances, a partial recovery.

Taken together, this analysis shows that groups of decision makers with diverse attention have the potential to outperform the average individual within the group. Exactly how much better the group performance is depends on the size of the group, the complexity of the problem space, and the size of individuals’ attention budget. The magnitude of this performance difference can be substantial, with groups of 10 being correct 10-20% more. Furthermore, the absolute group accuracy can be quite high. For example, a group of 10 decision makers with $\alpha=N1=3$ (i.e., they are using their full attention budget) is right 70% of the time in a problem space with $K=3$. This is 11% better than individuals.

The highest performing groups are those with all possible distinct low-dimensional representations. Having more, diverse representations is helpful relative to a randomly selected subset of all possible representations, but this might not be the optimal collection of representations. The cause of the decreasing returns to group size is that each additional representation is increasingly likely to make the same decisions as one or more of the representations already in the group. For example, if two models differ in their allocation of attention by only one dimension, their representations overlap (and have the same fitnesses) for $1/2$ of all perceived locations⁷. Thus, adding more decision-makers in an unstructured fashion is likely to increase the correlation of errors for a substantial number of decisions, which increases the probability that the group estimate will be wrong.

The optimal group of a given size would have the least correlated representations as possible. While identifying such a set of representations *a priori* would be challenging due to the natural variation in problem landscapes, organizations do introduce structure to the allocate the attention of their decision

⁷ $\frac{2^{N1-1}}{2^{N1}} = 1/2$

makers. They channel individuals' attention toward specific goals and tasks and this non-random structuring has the potential to enhance the predictive diversity and probability of decision-making success for groups of decision makers. I now turn to evaluating this possibility.

From Attentional Diversity to Predictive Diversity

The microstructural approach to understanding organizations argues that all organizations are built around a division of labor and then the integration of effort (Smith, 1776; March and Simon, 1993; Lawrence and Lorsch, 1967). A division of labor within an organization is generally viewed as necessary complication; by taking a holistic goal (e.g., profitability) and decomposing it into subgoals and tasks, an organization creates many problems of coordination that then need to be solved (Puranam, 2018). But just as that division creates efficiencies that allow organizations to outproduce individuals, here we ask if the same division can, in principle, produce the representational diversity that can allow a group of decision makers within an organization be a more accurate decision maker than individuals. The previous section demonstrated that diverse representations do enhance group decision-making, but the more important question is whether the allocation of attention that might be typical within organizations is better or worse than the randomly selected groups.

To begin to answer that question, I define an organization's *attention matrix*. An organization's "principle premises" (Simon, 1957) create its overarching goal, but labor and attention is divided out from there (Ocasio, 1997; Ocasio et al., 2021). An organization's attention matrix is a representation of that division in which the columns are dimensions of the problem space, and the rows are the decision-makers. The correspondence between the division of labor and the allocation of attention to dimensions need not be one-to-one. That is, the tasks and labor are not applied directly to the dimensions of the problem. I do assume, however, that the divisions generally track each other. For example, a semiconductor producing firm can have managers *cum* decision makers for research and development, production, the supply chain, sales, finance, and other functions. Within that division of labor, there are myriad distinct tasks (e.g., the design of chips, the sourcing of silica, the production of chips) that would not be a standalone dimension of the problem space but do map to the same dimension of the problem (e.g., the cost and purity of the silica). So, while the division of tasks might be complete (i.e., no two units would have the same task), different units or decision-makers might attend to the same dimensions of the problem space.

There is tremendous possible variety in the attention matrices. To make analysis tractable, I focus on five families that capture both features of theoretical interest and some of the possible variation. The precise matrices are shown in Figure 3, except for the first, the *full* matrix, in which all decision makers attend to all dimensions. This distribution of attention is antithetical to the very notion of selective attention, but provides an important theoretical baseline. At the other extreme is the *maximum divergence* matrix, in which attention is as divergent as possible, meaning that each decision maker

attends to as few dimensions as possible, with the constraint that each dimension is attended to by at least one decision-maker.

— INSERT FIGURE 3 ABOUT HERE —

In between these extremes are the *hierarchical*, *overlapping* and *incomplete* matrices. The *hierarchical* matrix captures a common way organizations might channel attention across the organization. At the top of the organization, the CEO attends to many dimensions of the problem space, while her subordinates attend to fewer. The hierarchical matrices used here do not have the CEO attending to all dimensions; instead there are some dimensions that only subordinates attend to, although each subordinate always attends to a dimension that his immediate superior also attends to, meaning there is some overlap in attention. The *overlapping* matrices exhibit this pattern as well, but each decision maker attends to the same number of dimensions. Finally, the *incomplete* matrices are hierarchical, but the organization fails to channel attention to the entire problem space. As Weick (2006) noted, the environment is an attentional construct insofar as we can individually or collectively fail to attend to its full extent, and the incomplete matrix will be used to explore this dimension of the distribution of attention.

Note that for these three types of matrix, for each value of M , the total amount of attention applied by the organization is held constant at an average of three dimensions per decision maker, or $\overline{N1} = 3$. This ensures that comparisons are exclusively about the distribution of attention, and not the availability of it. The total amount of attention does vary across M because having more decision makers attending to the same problem space is a key means of increasing total attention. Future work could consider the effect of varying the average $N1$.

Because it is the structure of these matrices that drives the difference between these groups and the randomly chosen ones, I characterize their *attentional diversity*, or how different each attention vector (a row of the matrix) is from all others. I formally measure this by looking at the specialization of attention. Following the logic of measures like term frequency-inverse document frequency (tf-idf) and the relative comparative advantage (Hidalgo and Hausmann, 2009). I define attentional diversity as one minus the average of the dot product of the column sums and the transpose of each row of the attention matrix, normalized by the product for the least diverse matrix (e.g., *full*). The column sums correspond to how much attention a dimension receives from all decision makers. The dot product of those sums with the transpose of decision maker's attention vector (row) increases as the decision maker attends to dimensions that receive more attention and represents how typical, relative to the group, a decision maker's attentional foci are. This typicality product is normalized to make it comparable across different values of M . To convert from an average typicality score to diversity, I make 1 minus the normalized average dot product the final measure. Table 1 contains the measure of attentional diversity for all matrices in Figure 3.

	Full	Incomplete	Hierarchical	Overlapping	Max Div.
M=5	0	.46	.673	.667	.8
M=10	0	.5	.683	.707	.9
M=20	0	.488	.667	.694	.9

Table 1: The attentional diversity of the matrices shown in Fig. 3.

When decision makers attend to the same dimensions, the attentional diversity is 0. The maximum value for each value of N and M corresponds to the *maximum divergence* matrix. The *incomplete* matrix is always the second least diverse and the *hierarchical* and *overlapping* are very similar. *Overlapping* is more diverse for two of three values of M . For the analyses below, results will be presented from least to most diverse.

To assess the relationship between attentional diversity and accuracy, I first apply the Diversity Prediction Theorem (Hong and Page, 2004) to all differences in fitnesses between all 2^{2N} pairs of locations in a landscape. According to the theorem, the group squared error on a prediction task is equal to the average individual squared error minus predictive diversity. For the two sides of the equation to be equal, predictive diversity must be the average error from the group estimate⁸. The average individual error, or *mean squared error*, can be large, and yet the group error can be small because the predictive diversity “cancels out” the individual errors. This ability to cancel out individual errors is the essence of beneficial cognitive diversity in groups.

For each pair of locations in a landscape, there is a true signed difference in the fitnesses and each decision-makers’ representation has a *perceived* signed difference. The average individual squared error for a given pair of points is the average of the squared difference between the true difference and the modeled difference for all decision makers. I use the average predicted difference to calculate predictive diversity; the squared difference between the average predicted difference and each individual’s predicted difference is itself averaged.⁹.

— INSERT FIGURE 4 ABOUT HERE —

Figure 4 captures the relationship between the group squared error, the mean squared error, and predictive diversity in graphical format for the attention matrices for $M = 10$. The figures for $M=5$ and 20 appear in the supplemental materials, as they are qualitatively similar. The results use 50 distinct sets of representations for each combination of K and α . That is, there are 50 problem spaces for each value of K , and for each value of α and the attention matrix type, a new set of representations

⁸Mathematically, this can be written as $(Y_t - C)^2 = \frac{\sum_{i=0}^n (Y_t - x_i)^2}{n} - \frac{\sum_{i=0}^n (C - x_i)^2}{n}$, where Y_t is the true value, x_i is the prediction of the i^{th} crowd member and $C = \frac{\sum_{i=0}^n x_i}{n}$.

⁹It is possible to calculate these quantities based on decisions rather than the estimate of the differences. However, doing so would require defining beforehand what the group decision-making method is and that is outside the scope of this paper.

is created for each of the 50 problem spaces. Averaging over 50 problem spaces avoids the possibility that the results are driven by a landscape with outlier characteristics. All values are normalized by the standard deviation of the fitnesses of the corresponding landscape because the variance of the fitnesses within an NK landscape decreases as K increases (Page, 2018), meaning the expected size of the error also decreases as a function of K .

Figure 4 exhibits several intuitive patterns. The more costly attention is (i.e., the lower α), the higher the mean squared error. The more complex the landscape (i.e., the higher K) is, again, the higher the mean squared-error. However, the predictive diversity is also highest in those regions, partially or almost completely offsetting the individual errors. As we should expect, when attention is free ($\alpha = 10$) and the attention matrix is *full*, the representations are fully accurate; there is no predictive diversity, but also no individual error. However, because of the existence of the attention budget, the full matrix is associated with the most errors overall. This suggests that decision-makers who attend to all dimensions of the problem and have limited attention available are going to perform poorly in any scenario; the individual errors are large, and there is little predictive diversity to save the group estimate.

There are also several non-obvious—but nonetheless sensible—patterns as well. When $K=0$ the landscapes are single-peaked, there is a substantial increase in accuracies relative to those with interdependencies, at least for lower values of α . This is the cause of the vertical bands on the left side of each panel. The horizontal bands at the bottom of many of the panels originate in the interaction between the distribution of attention and the budget. For example, when $\alpha = 0$, even decision makers embedded within a maximum divergence matrix will have errors. But when α increases to 9, because they each attend to one dimension, the size of the errors becomes constant. In the overlapping matrix, each decision maker attends to three dimensions, so the shift happens at $\alpha = 4$. This general pattern holds for both the hierarchical and incomplete matrices, but because some decision makers attend to more than others, the changes are less sharp.

When attention is costly, the *full* attention matrix results in highly inaccurate estimates. Much higher than the *maximum divergence* matrices, which do not exhibit much variation because individual attention is highly focused¹⁰. *Hierarchical* and *overlapping* matrices are almost identical; by shifting the allocation of attention from those at the top of the hierarchy down to others, the overlapping matrices make those at the top less accurate, but that loss is offset by the higher accuracy of others. Finally, the *incomplete* matrices are surprisingly accurate. A willingness to be uninformed in some areas in order to be more accurate in other areas is not a terrible strategy when the unattended areas are not many in number.

Taken together, there is an obvious relationship between attentional diversity, attention budget, and both the presence of predictive diversity and the size of individual errors. When attention is limited,

¹⁰When $K = 0$ (the problem space has a single peak) and attention is free, there is minimal error for the maximum divergence matrices, as predicted above.

distributed and diverse attention drives down individual errors while maintaining or even enhancing vital predictive diversity. This suggests that successful organizational decision-making is compatible, if not enhanced by the distribution of attention. As a final analysis, I therefore analyze the frequency with which groups make the correct decision as a crowd.

As in the first analysis, for the crowd estimate each decision-maker contributes an estimated difference between the two points being compared. The crowd estimate has a high potential for decision-making success; it stalls the conversion of judgments into decisions as long as possible. Contrast this with voting, which loses information insofar as individuals with wildly different estimates can nonetheless vote the same way (Csaszar and Eggers, 2013). The solicitation of crowd estimates is not common in organizations¹¹, but here serves as a crucial baseline for comparison because of the minimal amount of structure it imposes. It has the most potential to capitalize on the underlying cognitive diversity of the set of the decision makers. The crowd estimate, however, is not the same as the group squared error because even small errors can lead to the wrong decision.

— INSERT FIGURE 5 ABOUT HERE —

Figure 5 plots the performance results of the crowd estimate for all five attention matrices for $M=10$. The number of interdependencies, as controlled via the K parameter of the NK problem landscape, has a significant impact on the ability of the organization to make the correct decision. Figure 4 shows that once the attention budget is large enough, the size of the group squared error does not vary much as a function of K ; while the size of individual errors are increasing as the problem becomes more complex, the predictive diversity present in the set of decision-makers is increasing, thereby canceling out much of the individual error. Figure 5, however, reveals that the resulting small errors become increasingly likely to be in the wrong direction as K increases. This is why the surfaces slope down to the right. The only exception is for the *full* matrix, which does not vary in K . This is because those decision-makers have the same representations of the full problem space and the individual errors arise only through the existence of the attention budget.

Figure 5 also reveals both the benefits and limits of attentional diversity. For $K=0$, the performance of the crowd estimate increases monotonically as attentional, and predictive, diversity increases. But as K increases, a trade-off emerges; attentional diversity is most beneficial when attention is expensive (i.e., α is small), but the cheaper attention becomes, attentional diversity leads to worse performance. For example, the most diverse *overlapping* allocation of attention has high performance for $K=3$ and $\alpha=3$, about 15% better than the *hierarchical* matrix. However, for $K=6$ and $\alpha=6$, performance is about 5% worse. So while individually narrow, but collectively diverse attention can offset the loss of accuracy associated with having a small attention budget, as attention budgets expand the relationship

¹¹Although some organizations have run internal prediction markets that allow for the aggregation of point estimates (Cowgill and Zitzewitz, 2015). In private conversation, a key organizer of prediction markets at Google told the author that while the markets were highly successful, they were nonetheless shut down because they often embarrassed managers and undermined their authority.

reverses. This pattern holds for $M=5$ and $M=20$, the plots of which are relegated to the supporting information accordingly.

More generally, Figure 5 indicates that the structuring of attention within groups can lead to higher rates of correct decision-making than the randomly constituted groups. For example, a group embedded in an overlapping matrix is correct 77% of the time when given an attention budget of 3 and placed in an environment with $K=3$. This is better than the group of all 120 possible distinct representations (73%) and 10 randomly chosen representations (70%). However, when the group is embedded in the maximally divergent or hierarchical attention matrices, the rate is 63%. Although this might suggest that an overlapping attentional matrix is best, as noted above, this is conditional on the size of individuals' attention budgets and the complexity of the environment.

Discussion and Conclusion

To adjudicate the core claim that having diverse attentional foci can allow groups to outperform individuals, and generally perform well in decision-making tasks, the analysis above all compared the frequency with which individuals and groups of different constitutions can be expected to make the correct decisions. The overarching finding is that having diverse attentional foci allows groups to be substantially better decision makers than individuals. More attentionally diverse attention matrices can further enhance performance.

To get to that finding, the analysis first considered that frequency strictly as a function of the dimensionality of individuals' representation of the problem space. Intuitively, as individuals can attend to more dimensions, the percentage of correct decisions should increase. Within the model, this is true for both individuals and groups, but the groups have a higher percentage than the average individual for all values of $N1$, the number of dimensions attended to, and K , the strength of outcome interdependencies across dimensions. Furthermore, the larger the group, the better the performance.

There are, however, decreasing returns to the size of the group. This follows from the fact that adding representations that are uncorrelated with those already in the group gets harder the larger the group. The only exception to that pattern is when $K=0$ and $M=all$. In this case, the group is always perfectly rational. That is, in spite of having distinct low-dimensional representations of the space, the group's knowledge can be straightforwardly recombined to yield the correct answer. This is an important demonstration of how groups can, in principle, be collectively rational. However, there are at least two impediments to this ideal. First, the problem spaces organizations and groups are usually facing do not feature independence across dimensions. Figure 1 shows that as interdependencies are introduced, performance falls off substantially. But, second, there are not yet costs to spreading one's attention across more dimensions than is cognitively feasible if the goal is to gain meaningful insight into those dimensions.

To understand the role of such costs, I analyze the role of an attention budget, or parameter α .

Figure 2 documents that once the budget is smaller than the number of dimensions attended to, the performance declines. This follows directly from how the budget was implemented—when the budget is exceeded, the estimate of fitness of the perceived locations gets noisier—, but Figure 2 also shows that it is possible for it to be better to attend to more dimensions than the attention budgets allows full attention to. Particularly with groups, it is better to have noisier insights into more dimensions than more precise insights into fewer dimensions.

The introduction of organization-wide structure to individuals' attention, in the form of an attention matrix, provides further support for the claim that organizations can, in principle, be quite good decision-makers even when attention is individually sparse. The existence of predictive diversity, as documented in Figure 4, suggested that this would be the case. Figure 5 confirms that attentional diversity does translate to actual decision-making success.

The literature on attention and judgment and decision-making have homes in both psychology and the Carnegie approach to organizations, but the respective fields treat them quite differently. This is understandable given their different foci, but this paper attempts to bridge the gap by asking the question of whether the propensity of organizations to channel individuals' attention toward diverse aspects of the environment is, in fact, a driver of decision-making success.

The Carnegie notion of bounded rationality is rooted in the limited attentional and computational capacities of humans, and the attention-based view of the firm carries with it this pessimism (Porac and Tschang, 2013). To be sure, organizations are flawed systems of decision-making, but the psychology of attention and the wisdom of the crowd literature can offer a more optimistic take (Felin et al., 2017); while humans have limited attentional capacity, the mechanisms through which our attention is allocated often do a very good job extracting the most important information from the welter of stimuli we are subjected to (Posner and Rothbart, 2007; Hommel et al., 2019). Similarly, through its division of labor, a typical organization gives its members goals that encourage them to selectively apply their limited attention to their primary domain. If that focused attention makes them knowledgeable in their respective domains, the question is whether the low-dimensional representations decision-makers develop are useful when the organization needs to make a decision as a whole.

Drawing on the wisdom of the crowd literature, I argue that this is likely. Groups or crowds of people can be wise—that is, produce accurate collective judgments or predictions—when individual errors are canceled out by predictive diversity. Predictive diversity is simply a term for the distribution of errors around the group estimate, and much of the wisdom of the crowd literature is focused on understanding how to identify or cultivate the types of *cognitive diversity* that lead to predictive diversity. As I define it, attentional diversity is likely to engender cognitive diversity; limited attention leads to low-dimensional representations of the problem space that vary based only on the dimensions the individual attends to. When attention to the environment is distributed across members of the organization, the low-dimensional representations are likely to be predictively different.

I show that this representational or cognitive diversity translates into meaningful amounts of

predictive diversity. Predictive diversity is greatest when attention is most costly, but the size of individual errors is even larger. As attention moves from extremely costly to moderately costly, predictive diversity approaches the magnitude of the individual errors and the group estimate becomes highly accurate. This effect is stronger the more diverse the attention matrices are. Thus, when an organization can distribute attention in a diverse way, the individual errors tend to be symmetrically distributed around the true value and the group of decision makers has the potential to collectively be highly accurate.

While the maximum attentional diversity found in the *maximum divergence* matrix does create predictive diversity, it generally does not translate into decision-making success. Only when the decision-making environment does not have interdependencies (i.e., K is low) does this lead the matrix to a high probability of being correct. This reveals the limits of attentional diversity. To obtain the highest levels of attentional diversity, attention must become extremely narrow, and this narrowness leads to representations of the problem space that are unable to notice many differences. This inability to discriminate, while providing predictive diversity in a technical sense, undermines successful decision-making unless the environment is simple. So, while there is a linear relationship between attentional diversity and predictive diversity, there is a curvilinear relationship between attentional diversity and the probability of making the correct decision for a given attentional width.

One of the more interesting results is that the *incomplete* attentional matrix leads to surprisingly strong performance. In this matrix, four of the 10 dimensions are attended to by none of the decision makers, yet the decisions include proposals that cover all 10. This surprising robustness is accountable to both the nature of the decisions and predictive diversity. If, for example, a proposal changes five dimensions, in expectation only two are not attended to. The fitness contribution of those changes could be countervailing, and if not, there is still a good chance that the contributions from the changes in the other three attributes are larger. Any judgment errors could still be in the correct direction as well. This means that far fewer than 40% of decisions are wrong due to this attentional omission, as a first guess might suggest. Additionally, because no one is attending to those dimensions, the composition of individual errors in the group is unrelated to those missing attributes and predictive diversity is largely preserved (see Figure 4). This robustness is likely limited to organizations that fail to attend to fewer than half of all of the true number of dimensions. If more are ignored, the above logic gets reversed, and individual errors will be larger and more correlated.

Finally, as is well known in the Carnegie literature, interdependencies between dimensions make decision making harder, but here perhaps not as precipitously as expected. For example, decision makers with an overlapping attention matrix and an attention budget of $\alpha = 3$ are right 70% of the time for $K = 4$. Each decision maker is attending to three dimensions, and yet the fitness contribution of each dimension depends on the state of four other dimensions, meaning the decision maker cannot notice all relevant changes and the potential outcomes can be highly variable. Yet, collectively, decision accuracy does not fall off a cliff. The statistical relationships of the N landscape are still

largely opaque (although see Herrmann et al. (2016)), as are the statistics of the low-dimensional representations used here, but the success observed here merits further study.

While the primary goal of this paper is to establish a framework around which the implications of diverse attention can be related to organizational decision-making and not to make concrete claims about organizational performance, the model strongly suggests that diverse attention can be compatible with successful decision-making under a range of circumstances. However, the circumstances considered here are only the allocation and cost of attention, and the structure of the organization's decision-making environment. Decades of research within the Carnegie tradition and psychology can point to a host of other factors that would limit the probability of decision-making success. Crucially, the model ignores any issues around goal alignment or competitive political coalitions (Cyert and March, 1992). The interests of the agents do not conflict with their principles or peers, so there is no issue of agents withholding information or making the recommendation that they believe would personally benefit them more. Yet, this is a real problem and one that is related the distribution of attention; attention and goal setting are tightly linked (Dijksterhuis and Aarts, 2010; Chun et al., 2011) and one of the ways organizations successfully channel attention is through the establishment of more proximate goals for decision makers (Ocasio, 2011). Managers can hope that the proximate goals align with the distal goal of overall organizational performance, but often they do not. Thus a decision-maker who has reason to believe that a proposed change will harm performance related to their proximate goal while improving overall organizational performance might well share the opposite of their true judgment.

Also, while the model of low-dimensional representations has surface validity, there are non-attentional sources of cognitive diversity that could impact judgments, including cultural, economic, and educational contexts. A range of evaluative biases could further skew representations. In defining the model, I offered some reasons to believe that having diverse foci of attention would prevent these considerations from creating the types of correlations across representations that would undercut the conclusions of the model, but this has not been tested.

These realities do not render this modeling exercise meaningless; rather, the model begins to contextualize this issue and others. It shows that the allocation of attention across organizations is a meaningfully manipulable part of organizational design that also contributes to the explanation of when and why organizations perform better than random chance and worse than full rationality. The existence of these bounds can inform the study of other aspects of individual and organizational decision-making (Siggelkow, 2011; Knudsen et al., 2019).

Attentional processes are a core element of individual cognition and decision-making. What we attend to informs our understanding of the situation and how we evaluate alternatives. The Carnegie approach views organizations as crucial external determinants of how individuals allocate attention and make decisions. This paper moves from the level of the individual to the level of the whole organization to explore the properties of an organization as a system of distributed attention. If

individuals cannot attend to all salient features of the environment, is there any promise in a system that deploys different attentional subunits? If one views organizations as information processing systems, which the Carnegie approach very much does (Joseph and Gaba, 2020), using multiple attentional subunits parallelizes computation in a way individuals cannot. As Moray (1967) noted, parallelization can get around limited channel or processing capacity. But this is not true parallelization because the signals being processed are not independent. The question thus becomes whether attention-driven partial processing of the signal can be meaningfully reintegrated or if parallel processing results in too much loss. If it can be reintegrated, it is part of the answer to Stinchcombe’s demand that any theory of organizations needs to account for “how organizations can be more rational than individuals (though of course they are not always)” [p. 340](Stinchcombe, 1990).

This model shows that it can be meaningfully reintegrated. I argue that rather than turning to social influence and the deliberation to reintegrate, an organization can rely on the principles of the “wisdom of the crowd.” When a group of individuals produces diverse predictions about an outcome, individual errors can (partially) cancel each other out so that the group estimate is highly accurate. The literature on the wisdom of the crowd argues that a primary source of predictive diversity is cognitive diversity, and I, in turn, argue that attentional diversity can be a source of cognitive diversity. So when an organization allocates attention diversely, it is creating conditions ripe for wise crowds. The model shows that this can be true, although the ability to translate this beneficial predictive diversity into correct decisions depends on the decision-making process and how many interdependencies exist in the environment.

To the best of my knowledge, this result is new to the psychology of attention and group decision-making. An unstructured group will likely not have the amount of attentional diversity to get a strong wisdom of the crowd effect, but structuring groups as organizations do can enhance outcomes. The collective intelligence literature has established c or ci —complementing g , the statistical factor of general intelligence—as a measure of how much a group outperforms a collection of individuals (Woolley et al., 2010; Riedl et al., 2021) and the intentional structuring of attention has the potential to enhance c . This ability to design organizations to enhance performance is also of interest to Carnegie researchers and a point around which psychologists and organizational scholars can deepen dialogues and collaborations.

Data Availability Statement

The code for the model and the experiment are available at [NAME OF REPOSITORY TO COME]. The experiment produced about 28GBs of data. Because of the amount of it, the data are not available in a repository but are available on request from the author.

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Main Figures

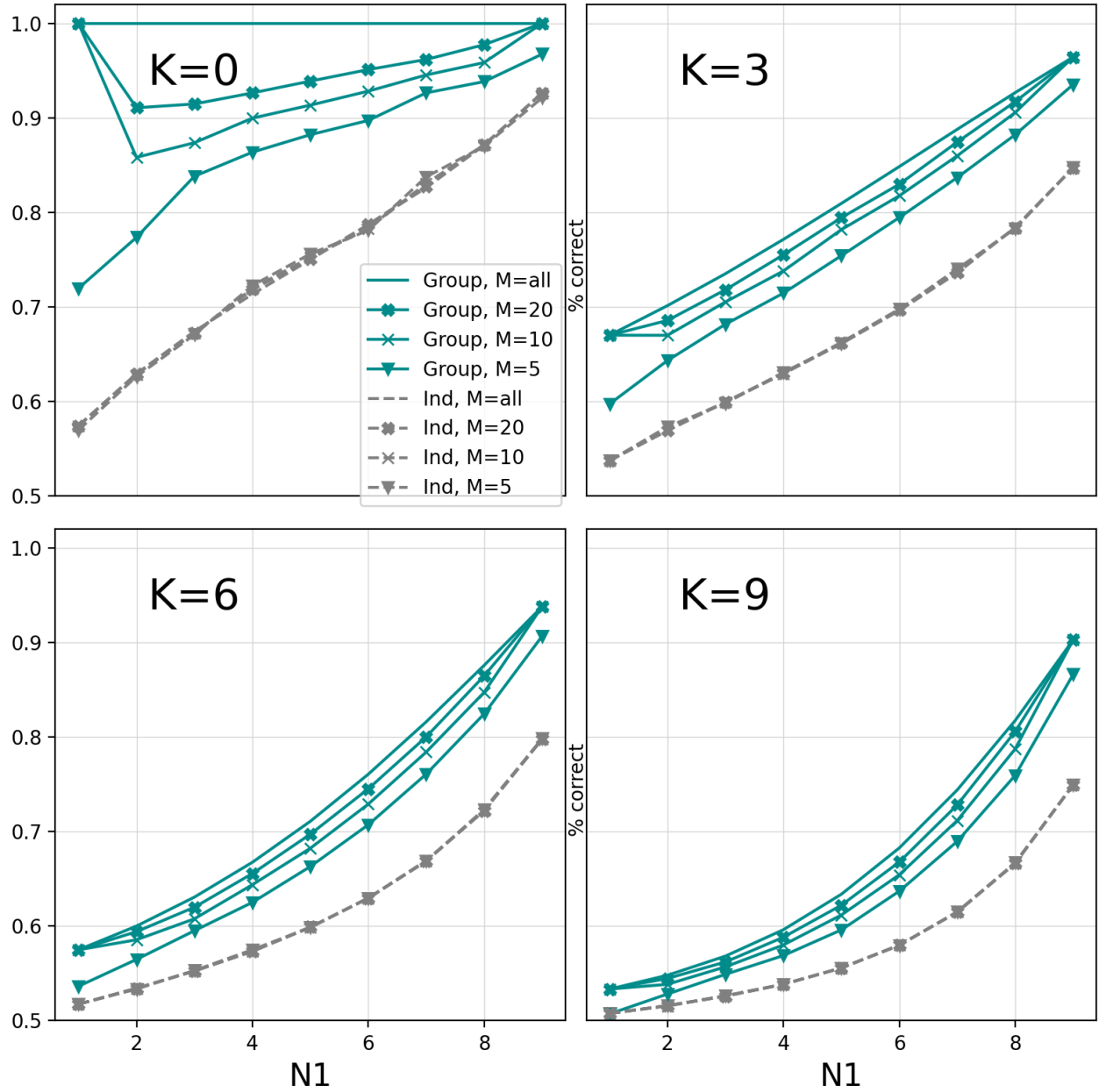


Figure 1: The effect of attention span ($N1$) on accuracy. The panels correspond to different levels of problem interdependency (i.e., K) and show the average accuracy, or percentage of decisions that are correct, for both individual and groups. For individuals, each point is the average percentage of correct decisions for different random sets of unique low-dimensional representations (i.e., M), for 50 distinct landscapes for the given value of K . For groups, the M decision-makers contribute to a single group decision for each comparison so each point is the average across 50 landscapes. All possible 1,048,576 comparisons or decisions are considered.

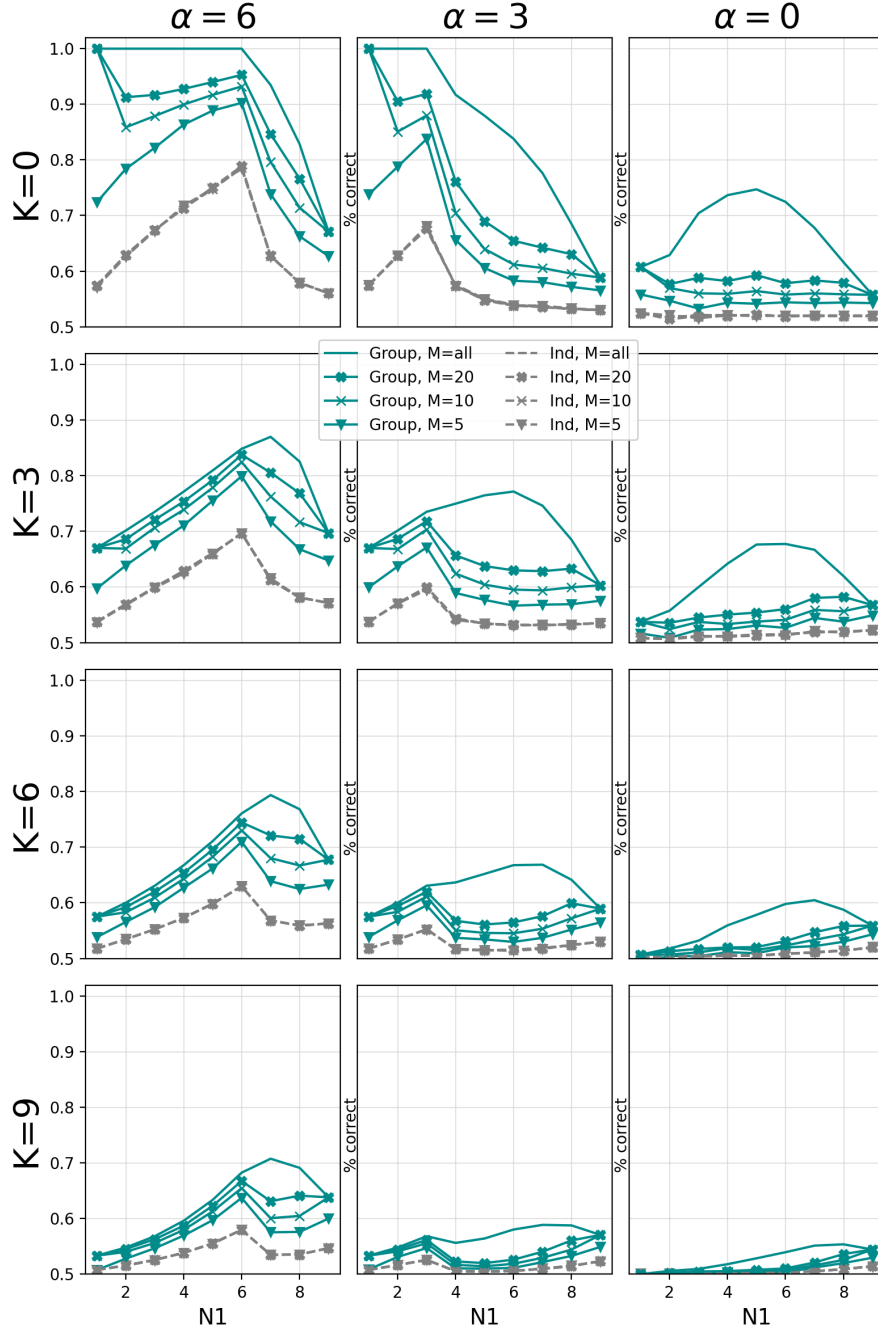


Figure 2: The effect of attention budget on accuracy. The columns vary the size of the budget α and the rows vary the levels of problem interdependency (i.e., K). Each panel shows the average accuracy, or percentage of decisions that are correct, for both individual and groups. For individuals, each point is the average percentage of correct decisions for different random sets of unique low-dimensional representations (i.e., M), for 50 distinct landscapes for the given value of K . For groups, the M decision-makers contribute to a single group decision for each comparison so each point is the average across 50 landscapes. All possible 1,048,576 comparisons or decisions are considered.

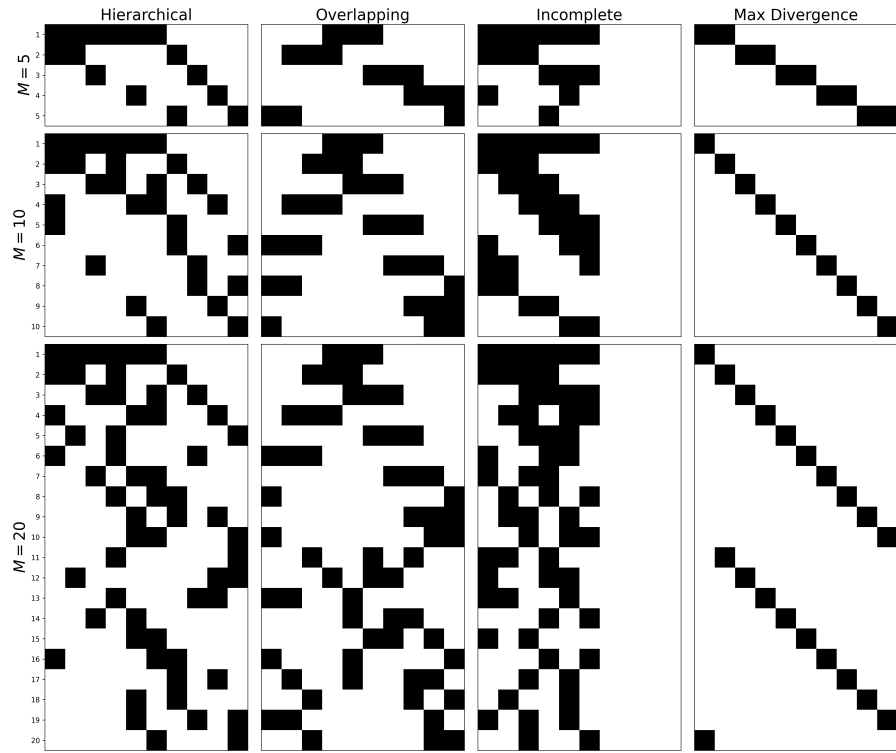


Figure 3: Attention matrices. The precise attention matrices used in the second analysis. The *full* matrices are omitted.

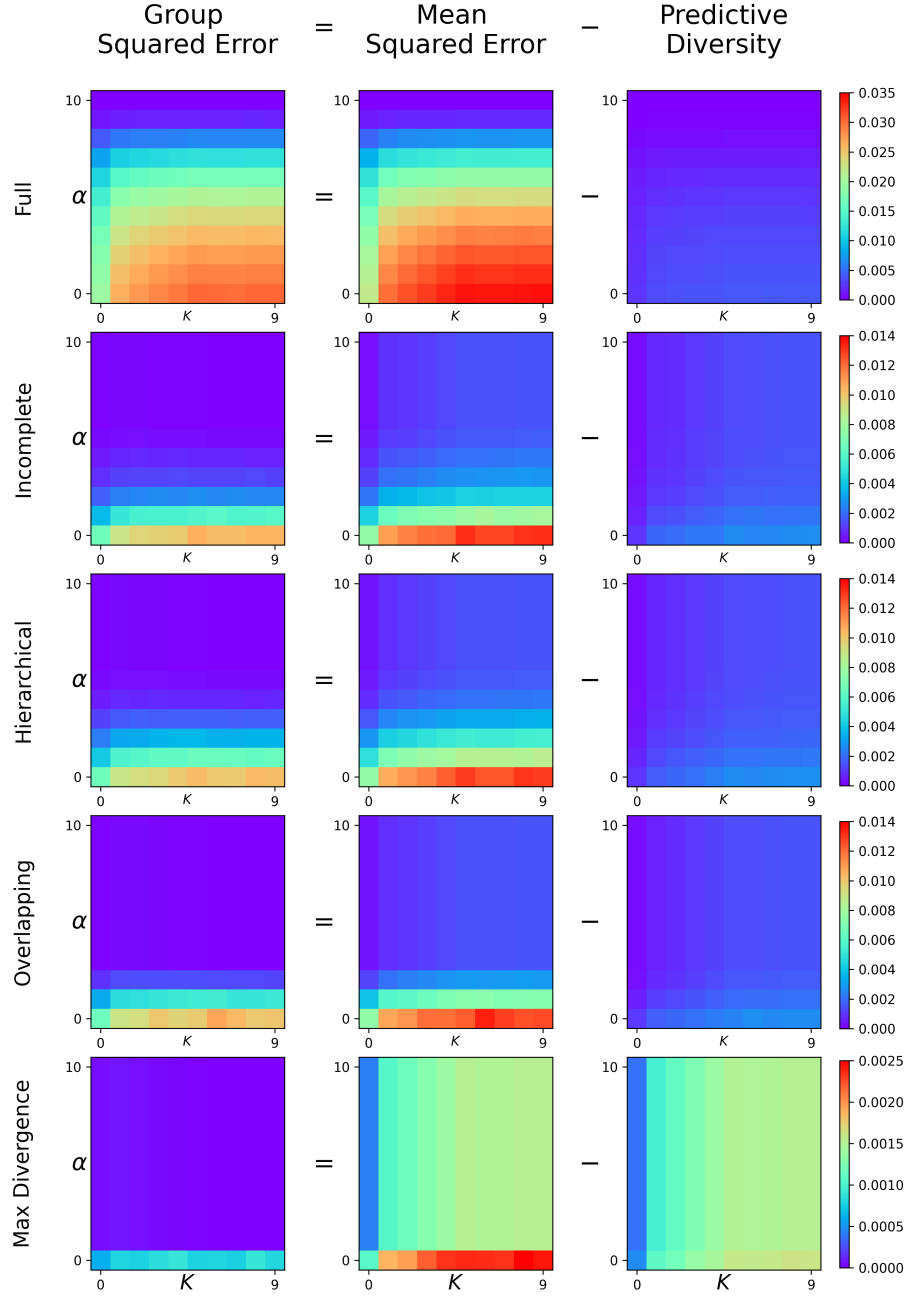


Figure 4: The Diversity Prediction Theorem in graphical form. The rows are associated with different attention matrices for $M = 10$ and the columns are the group squared error, the mean squared error and predictive diversity, respectively. The panel in each cell plots the corresponding values for different values of K and α . The values are normalized by the standard deviation of the associated landscape because the variances in fitness decreases in K .

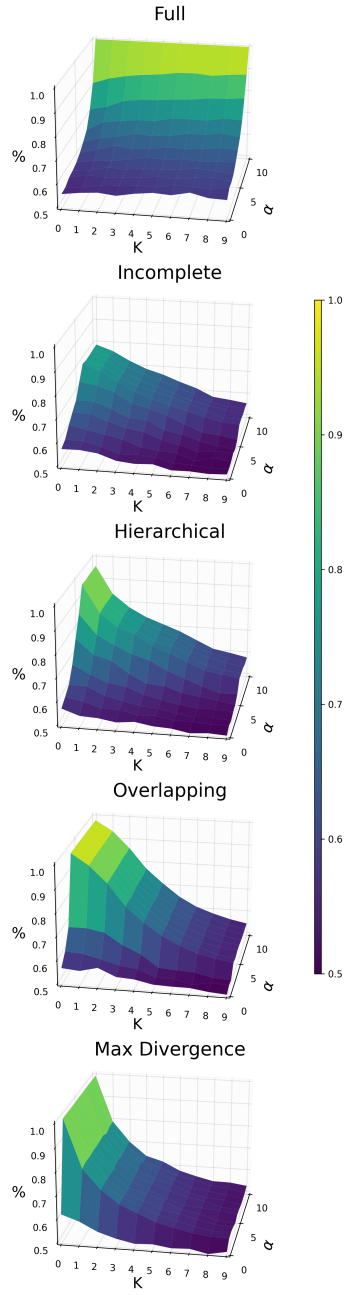


Figure 5: Color mapped surfaces depicting the performance of the crowd estimate over different values of K and α for $M = 10$. The attention matrices are ordered from lowest to highest attentional diversity. The z-axis of each plot is the percentage of correct decisions for the crowd estimate.

Supplemental Figures

M=5 figures

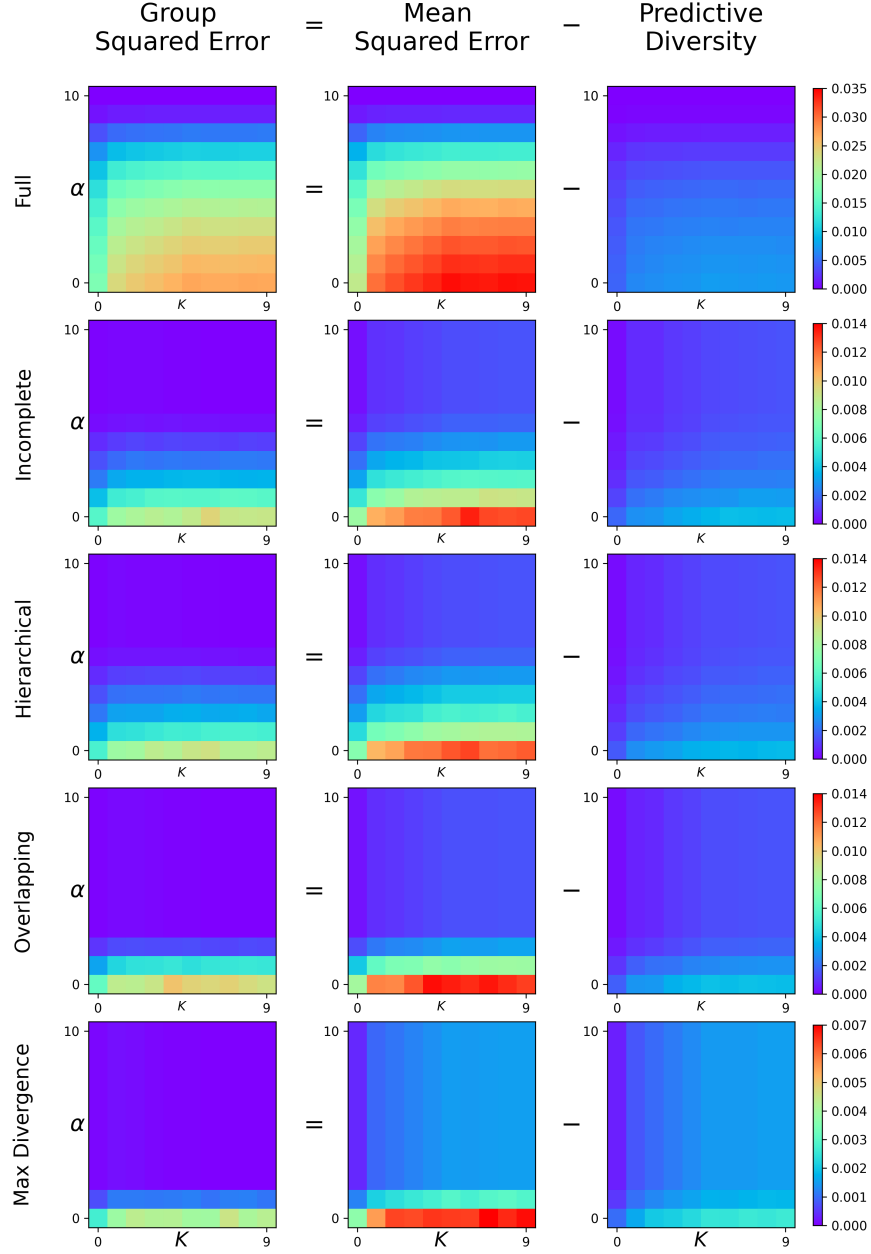


Figure 1: The Diversity Prediction Theorem in graphical form. The rows are associated with different attention matrices for $M = 5$ and the columns are the group squared error, the mean squared error and predictive diversity, respectively. The panel in each cell plots the corresponding values for different values of K and α . The values are normalized by the standard deviation of the associated landscape because the variances in fitness decreases in K .

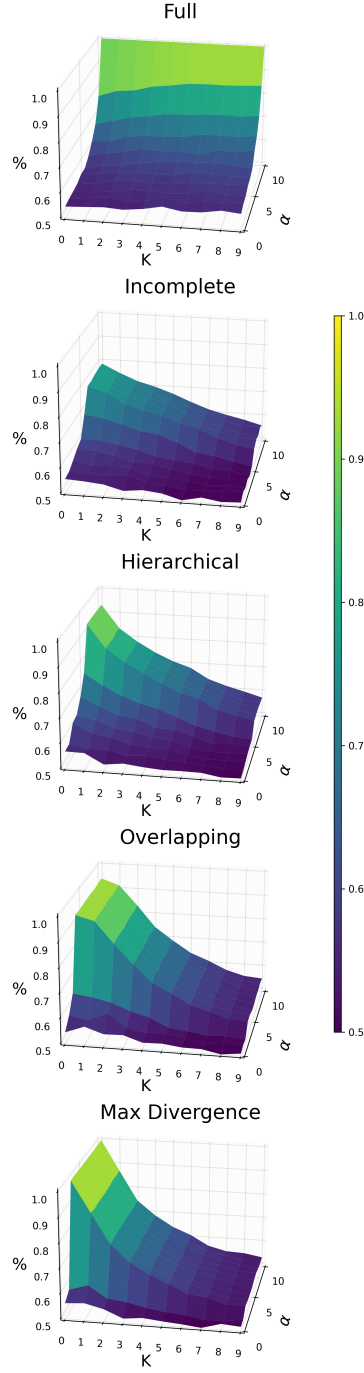


Figure 2: Color mapped surfaces depicting the performance of the crowd estimate over different values of K and α for $M = 5$. The attention matrices are ordered from lowest to highest attentional diversity. The z-axis of each plot is the percentage of correct decisions for the crowd estimate.

M=20 figures

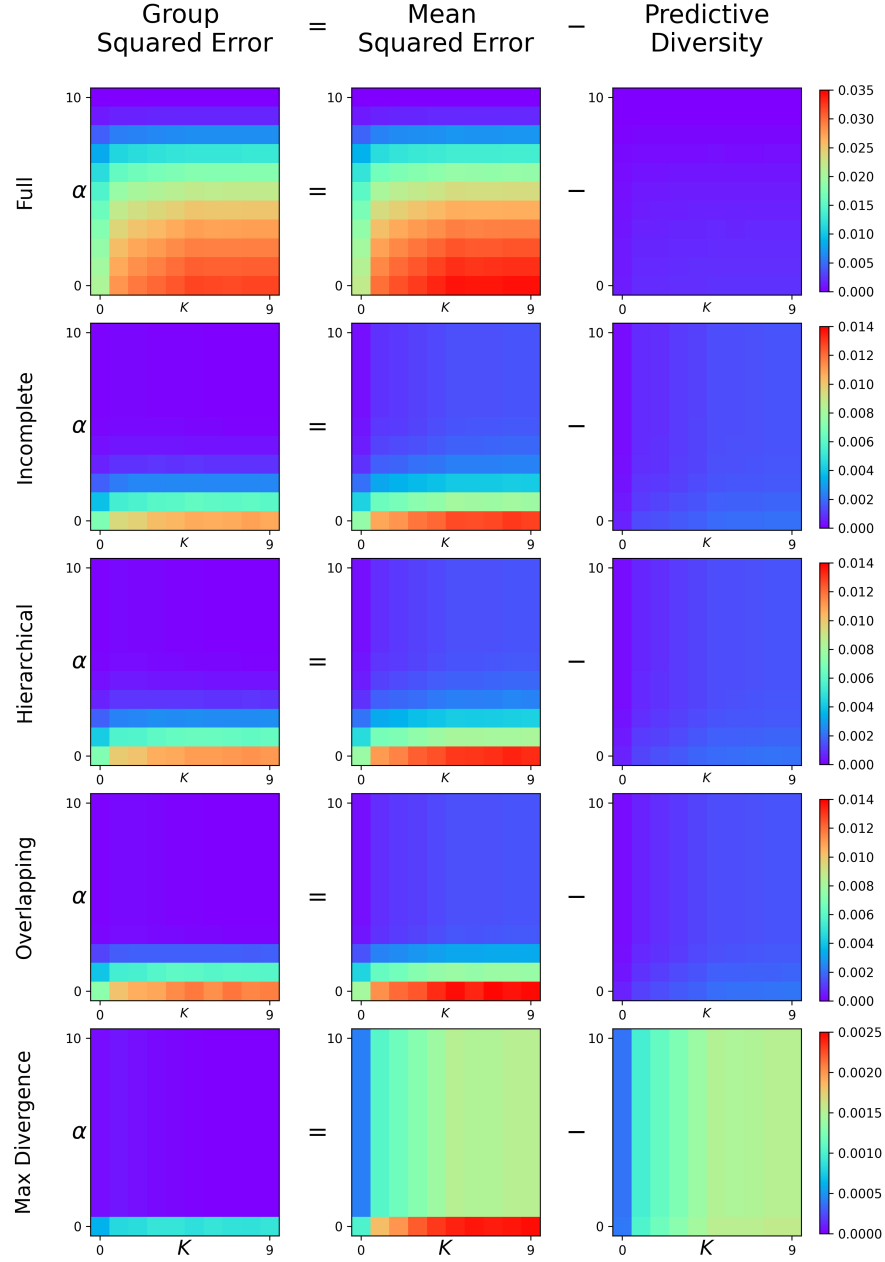


Figure 3: The Diversity Prediction Theorem in graphical form. The rows are associated with different attention matrices for $M = 20$ and the columns are the group squared error, the mean squared error and predictive diversity, respectively. The panel in each cell plots the corresponding values for different values of K and α . The values are normalized by the standard deviation of the associated landscape because the variances in fitness decreases in K .

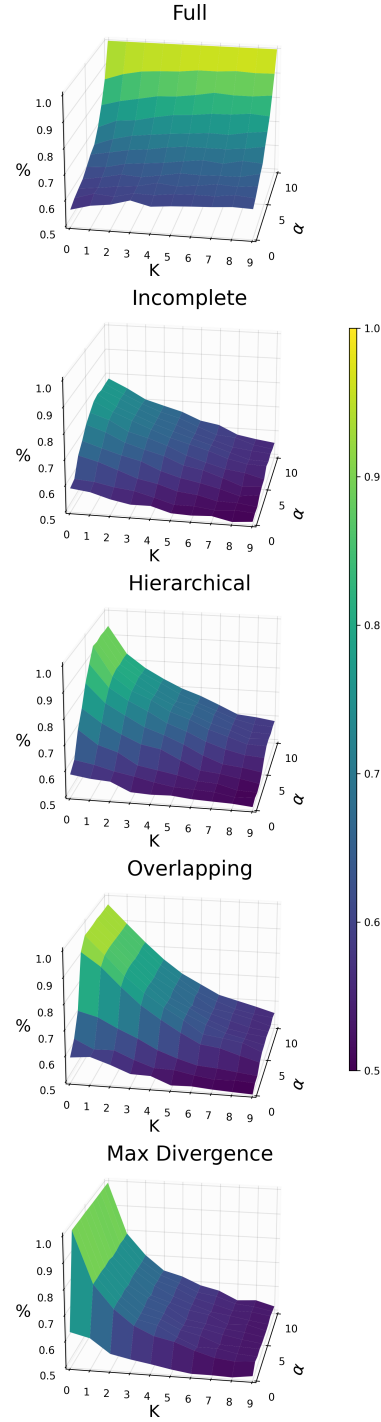


Figure 4: Color mapped surfaces depicting the performance of the crowd estimate over different values of K and α for $M = 20$. The attention matrices are ordered from lowest to highest attentional diversity. The z-axis of each plot is the percentage of correct decisions for the crowd estimate.